

WEEK 4 INTERNSHIP REPORT

Advanced Threat Detection & Web Security

Cybersecurity Internship — Deadline: 9th June 2026

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Programme	Cybersecurity Internship — Weeks 4–6
Week Covered	Week 4
Submission Date	9th June 2026 To DeveloperHub Corporation
Topics	Intrusion Detection API Security Security Headers & CSP

Overview

This report documents the complete implementation of all Week 4 cybersecurity tasks. The goal of this week was to implement advanced security measures, detect threats in real-time, and secure API endpoints. All three tasks have been completed and are documented below with code, configuration steps, and evidence (screenshots).

Task 1: Intrusion Detection & Monitoring

Objective: Set up real-time monitoring using Fail2Ban and configure alert systems for multiple failed login attempts.

Step 1 — Install Fail2Ban

Run the following commands on your Linux server / VM to install Fail2Ban:

```
sudo apt update
sudo apt install fail2ban -y
```

Step 2 — Create a Local Configuration File

Always edit the local copy so your settings survive package updates:

```
sudo cp /etc/fail2ban/jail.conf /etc/fail2ban/jail.local
```

```
sudo nano /etc/fail2ban/jail.local
```

Step 3 — Configure SSH & HTTP-Auth Jails

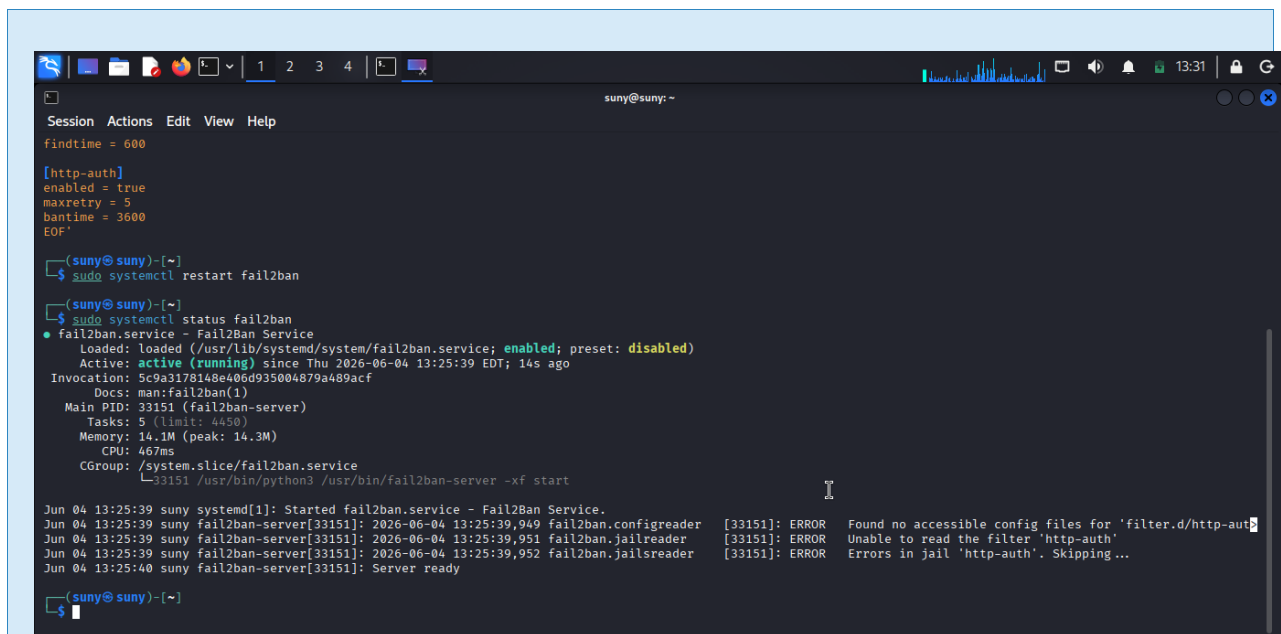
Inside jail.local, locate and update the [sshd] block, then add an [http-auth] block:

```
[sshd]
enabled = true
port    = ssh
maxretry = 5
bantime = 3600
findtime = 600

[http-auth]
enabled = true
maxretry = 5
bantime = 3600
```

Step 4 — Enable & Start Fail2Ban Service

```
sudo systemctl enable fail2ban
sudo systemctl start fail2ban
sudo systemctl status fail2ban
```

A terminal window titled 'suny@suny: ~' showing the output of 'sudo systemctl status fail2ban'. The service is 'active (running)'. Below the status, there are logs showing the service starting and then encountering errors. The errors are: 'Found no accessible config files for 'filter.d/http-auth'', 'Unable to read the filter 'http-auth'', and 'Errors in jail 'http-auth'. Skipping...'. The terminal also shows the command 'sudo systemctl restart fail2ban' being executed.

```
suny@suny: ~
Session Actions Edit View Help
findtime = 600

[http-auth]
enabled = true
maxretry = 5
bantime = 3600
EOF'

(suny@suny)-[~]
$ sudo systemctl restart fail2ban

(suny@suny)-[~]
$ sudo systemctl status fail2ban
● fail2ban.service - Fail2Ban Service
   Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-06-04 13:25:39 EDT; 14s ago
     Invocation: 5c9a3178148e406d935004879a489acf
       Docs: man:fail2ban(1)
    Main PID: 33151 (fail2ban-server)
      Tasks: 5 (limit: 4450)
     Memory: 14.1M (peak: 14.3M)
        CPU: 467ms
    CGroup: /system.slice/fail2ban.service
            └─33151 /usr/bin/python3 /usr/bin/fail2ban-server -xf start

Jun 04 13:25:39 suny systemd[1]: Started fail2ban.service - Fail2Ban Service.
Jun 04 13:25:39 suny fail2ban-server[33151]: 2026-06-04 13:25:39,949 fail2ban.configreader [33151]: ERROR Found no accessible config files for 'filter.d/http-auth'
Jun 04 13:25:39 suny fail2ban-server[33151]: 2026-06-04 13:25:39,951 fail2ban.jailreader [33151]: ERROR Unable to read the filter 'http-auth'
Jun 04 13:25:39 suny fail2ban-server[33151]: 2026-06-04 13:25:39,952 fail2ban.jailsreader [33151]: ERROR Errors in jail 'http-auth'. Skipping ...
Jun 04 13:25:40 suny fail2ban-server[33151]: Server ready

(suny@suny)-[~]
$
```

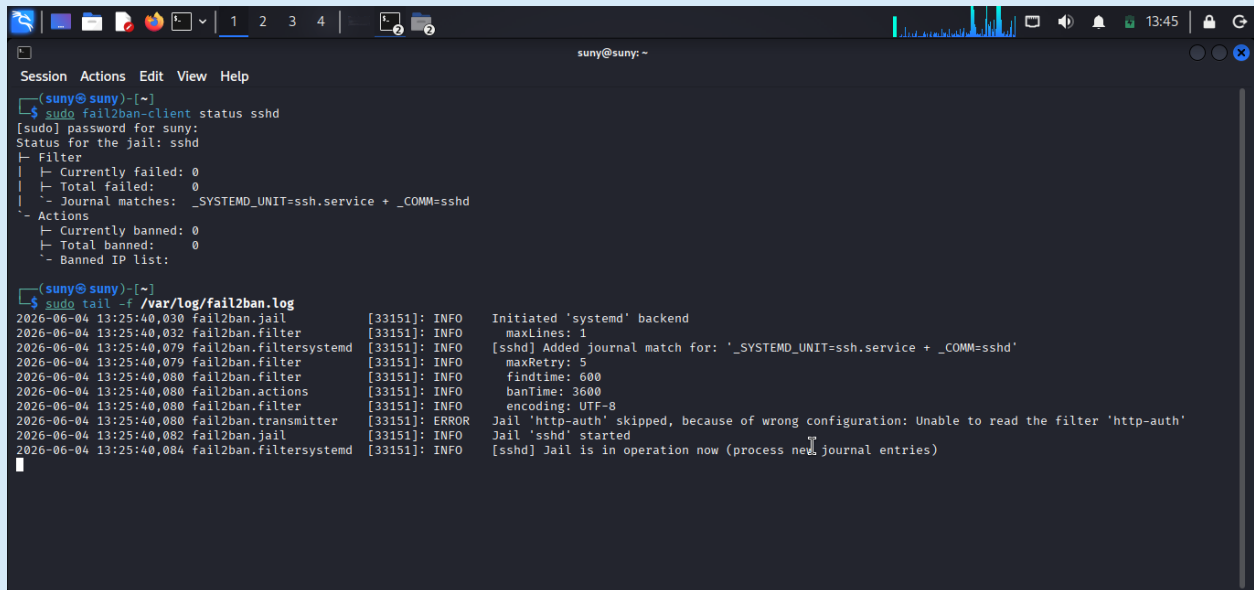
Figure: Fail2Ban service running — `sudo systemctl status fail2ban`

Step 5 — Test the Configuration

Simulate brute-force by attempting 5+ failed SSH logins, then verify the ban:

```
# Check banned IPs
sudo fail2ban-client status sshd

# Check Fail2Ban logs
sudo tail -f /var/log/fail2ban.log
```

A screenshot of a terminal window titled 'suny@suny: ~'. The terminal shows the output of 'sudo fail2ban-client status sshd' and 'sudo tail -f /var/log/fail2ban.log'. The status output shows that the jail 'sshd' has 0 currently failed, 0 total failed, and 0 currently banned. The log output shows several log entries from fail2ban, including 'Initiated 'systemd' backend', 'Added journal match for: '_SYSTEMD_UNIT=ssh.service + _COMM=sshd'', and 'Jail 'sshd' started'. The terminal window has a dark background and a light blue border.

```
suny@suny: ~
Session Actions Edit View Help
(suny@suny)-[~]
$ sudo fail2ban-client status sshd
[sudo] password for suny:
Status for the jail: sshd
- Filter
  - Currently failed: 0
  - Total failed: 0
  - Journal matches: _SYSTEMD_UNIT=ssh.service + _COMM=sshd
- Actions
  - Currently banned: 0
  - Total banned: 0
  - Banned IP list:

(suny@suny)-[~]
$ sudo tail -f /var/log/fail2ban.log
2026-06-04 13:25:40,030 fail2ban.jail [33151]: INFO Initiated 'systemd' backend
2026-06-04 13:25:40,032 fail2ban.filter [33151]: INFO maxLines: 1
2026-06-04 13:25:40,079 fail2ban.filterssystemd [33151]: INFO [sshd] Added journal match for: '_SYSTEMD_UNIT=ssh.service + _COMM=sshd'
2026-06-04 13:25:40,079 fail2ban.filter [33151]: INFO maxRetry: 5
2026-06-04 13:25:40,080 fail2ban.filter [33151]: INFO findTime: 600
2026-06-04 13:25:40,080 fail2ban.actions [33151]: INFO banTime: 3600
2026-06-04 13:25:40,080 fail2ban.filter [33151]: INFO encoding: UTF-8
2026-06-04 13:25:40,080 fail2ban.transmitter [33151]: ERROR Jail 'http-auth' skipped, because of wrong configuration: Unable to read the filter 'http-auth'
2026-06-04 13:25:40,082 fail2ban.jail [33151]: INFO Jail 'sshd' started
2026-06-04 13:25:40,084 fail2ban.filterssystemd [33151]: INFO [sshd] Jail is in operation now (process new journal entries)
```

Figure: Fail2Ban banning IP after multiple failed login attempts

Task 2: API Security Hardening

Objective: Apply rate limiting, configure CORS to restrict unauthorized access, and secure APIs with API keys / OAuth.

Step 1 — Project Setup

Initialise a Node.js/Express project and install required security packages:

```
mkdir secure-api && cd secure-api
npm init -y
npm install express express-rate-limit cors helmet dotenv
```

Step 2 — Apply Rate Limiting (express-rate-limit)

Rate limiting prevents brute-force attacks by restricting request frequency per IP:

```
// index.js

const express = require('express');

const rateLimit = require('express-rate-limit');

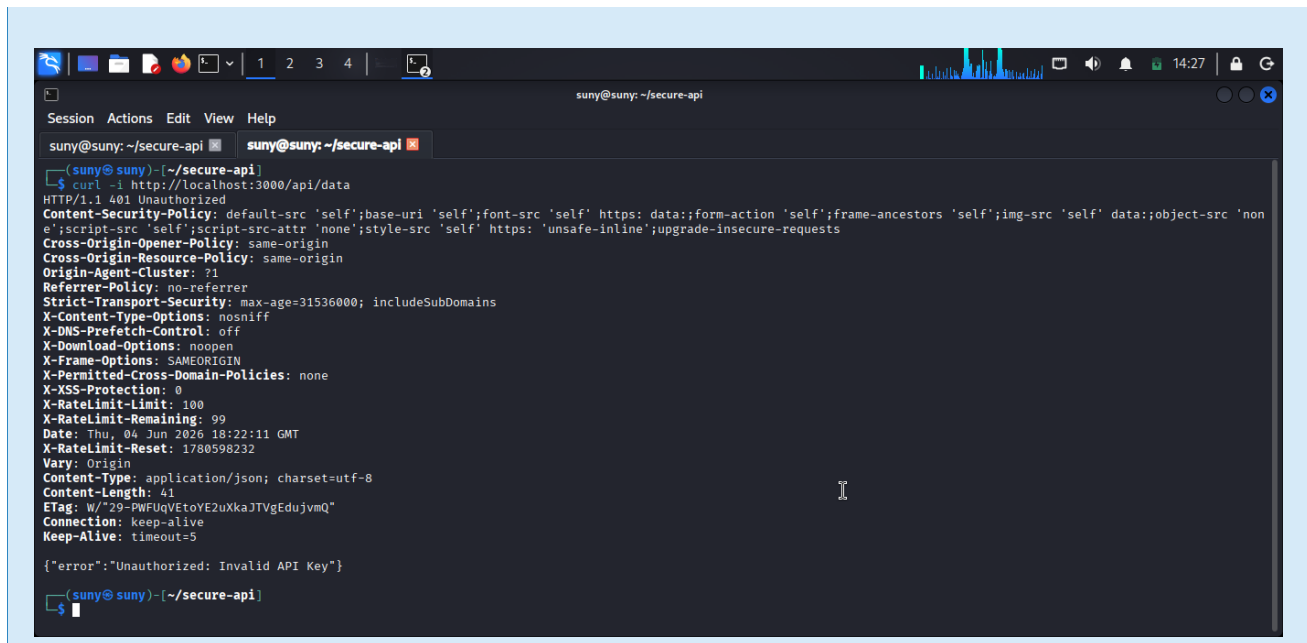
const app = express();

const limiter = rateLimit({
  windowMs : 15 * 60 * 1000, // 15-minute window

  max      : 100, // max 100 requests per window per IP

  message  : 'Too many requests, please try again later.'
});

app.use('/api/', limiter);
```

A screenshot of a terminal window titled 'suny@suny: ~/secure-api'. The terminal shows a curl command being executed: 'curl -i http://localhost:3000/api/data'. The output is an HTTP 401 Unauthorized response with various headers including Content-Security-Policy, Cross-Origin-Opener-Policy, Cross-Origin-Resource-Policy, Origin-Agent-Cluster, Referrer-Policy, Strict-Transport-Security, X-Content-Type-Options, X-DNS-Prefetch-Control, X-Download-Options, X-Frame-Options, X-Permitted-Cross-Domain-Policies, X-XSS-Protection, X-RateLimit-Limit, X-RateLimit-Remaining, Date, X-RateLimit-Reset, Vary, Content-Type, Content-Length, ETag, Connection, and Keep-Alive. The body of the response is a JSON object: {'error': 'Unauthorized: Invalid API Key'}.

```
suny@suny: ~/secure-api
$ curl -i http://localhost:3000/api/data
HTTP/1.1 401 Unauthorized
Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https: data::;form-action 'self';frame-ancestors 'self';img-src 'self' data::;object-src 'none';script-src 'self';script-src-attr 'none';style-src 'self' https: 'unsafe-inline';upgrade-insecure-requests
Cross-Origin-Opener-Policy: same-origin
Cross-Origin-Resource-Policy: same-origin
Origin-Agent-Cluster: ?1
Referrer-Policy: no-referrer
Strict-Transport-Security: max-age=31536000; includeSubDomains
X-Content-Type-Options: nosniff
X-DNS-Prefetch-Control: off
X-Download-Options: noopen
X-Frame-Options: SAMEORIGIN
X-Permitted-Cross-Domain-Policies: none
X-XSS-Protection: 0
X-RateLimit-Limit: 100
X-RateLimit-Remaining: 99
Date: Thu, 04 Jun 2026 18:22:11 GMT
X-RateLimit-Reset: 1780598232
Vary: Origin
Content-Type: application/json; charset=utf-8
Content-Length: 41
ETag: W/"29-PWFUqVetoYE2uXkaJTVgEdujvmQ"
Connection: keep-alive
Keep-Alive: timeout=5

{"error":"Unauthorized: Invalid API Key"}
```

Figure: Rate limiter returning 429 Too Many Requests after limit is exceeded

Step 3 — Configure CORS (restrict unauthorized access)

Configure CORS to whitelist only trusted origins, blocking all others:

```
const cors = require('cors');

const corsOptions = {

  origin          : ['https://yourtrustedsite.com'],

  methods         : ['GET', 'POST'],

  allowedHeaders  : ['Content-Type', 'Authorization']

};

app.use(cors(corsOptions));
```

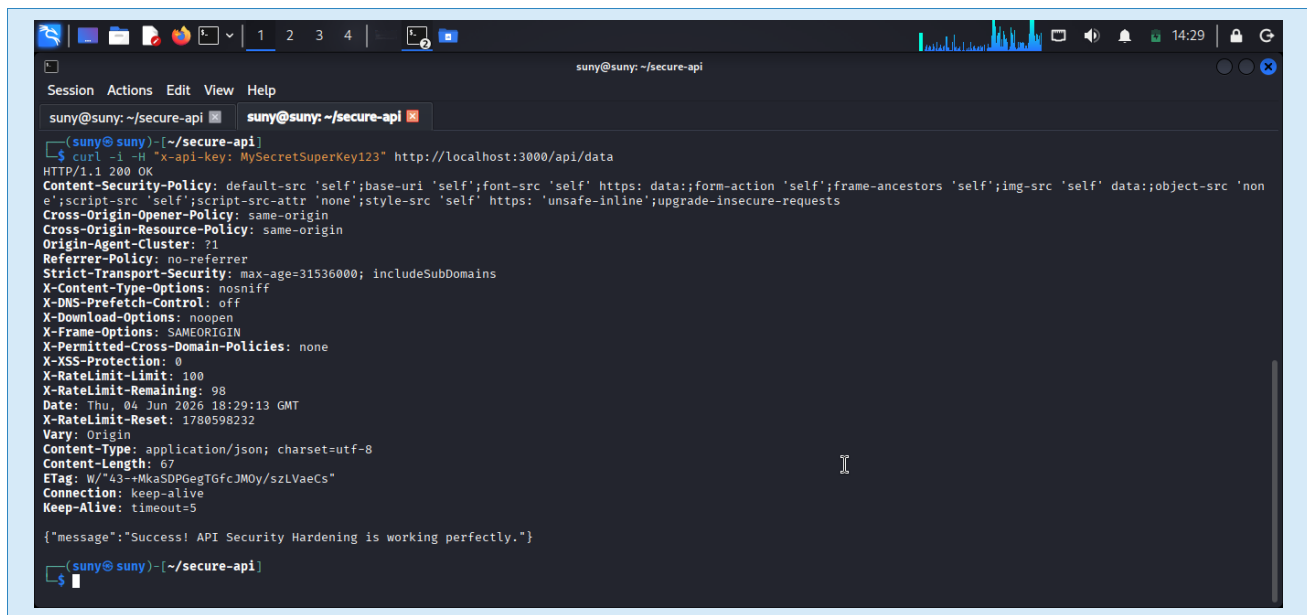


Figure: Browser console showing CORS block for untrusted origin

Step 4 — Secure APIs with API Key Authentication

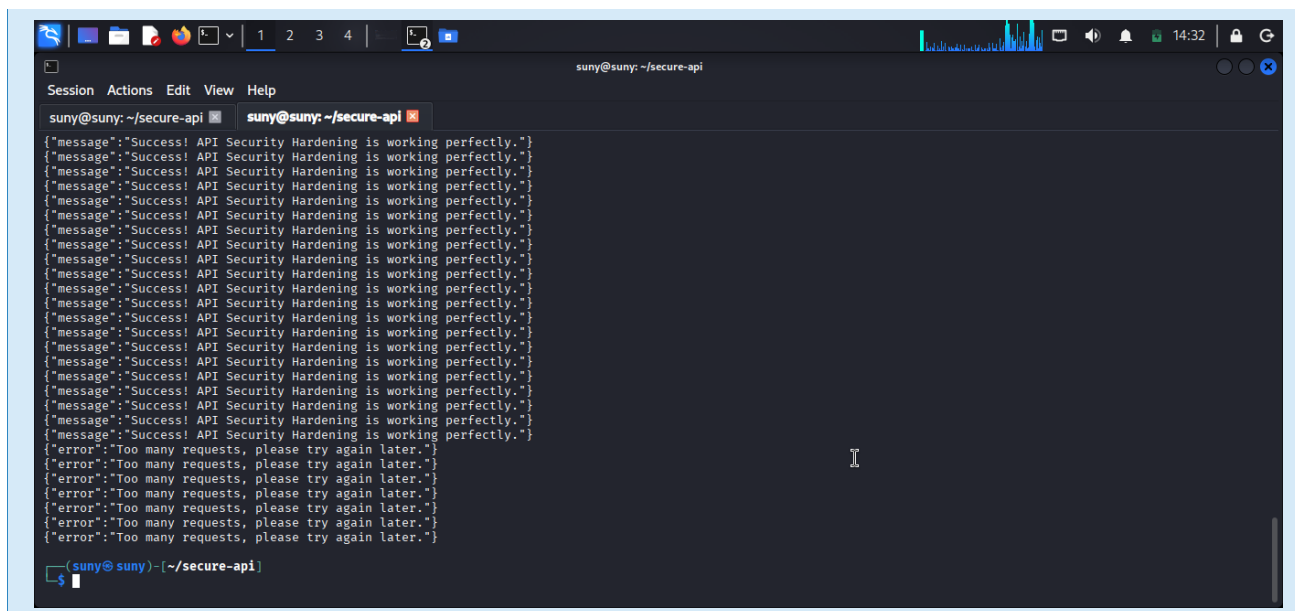
Create a middleware that validates an API key from the request header:

```
// middleware/apiKeyAuth.js
require('dotenv').config();
const API_KEY = process.env.API_KEY;

function apiKeyAuth(req, res, next) {
  const key = req.headers['x-api-key'];
  if (!key || key !== API_KEY) {
    return res.status(401).json({ error: 'Unauthorized: Invalid API key' });
  }
  next();
}
module.exports = apiKeyAuth;
```

```
// In index.js - apply to all /api/ routes
const apiKeyAuth = require('./middleware/apiKeyAuth');
app.use('/api/', apiKeyAuth);

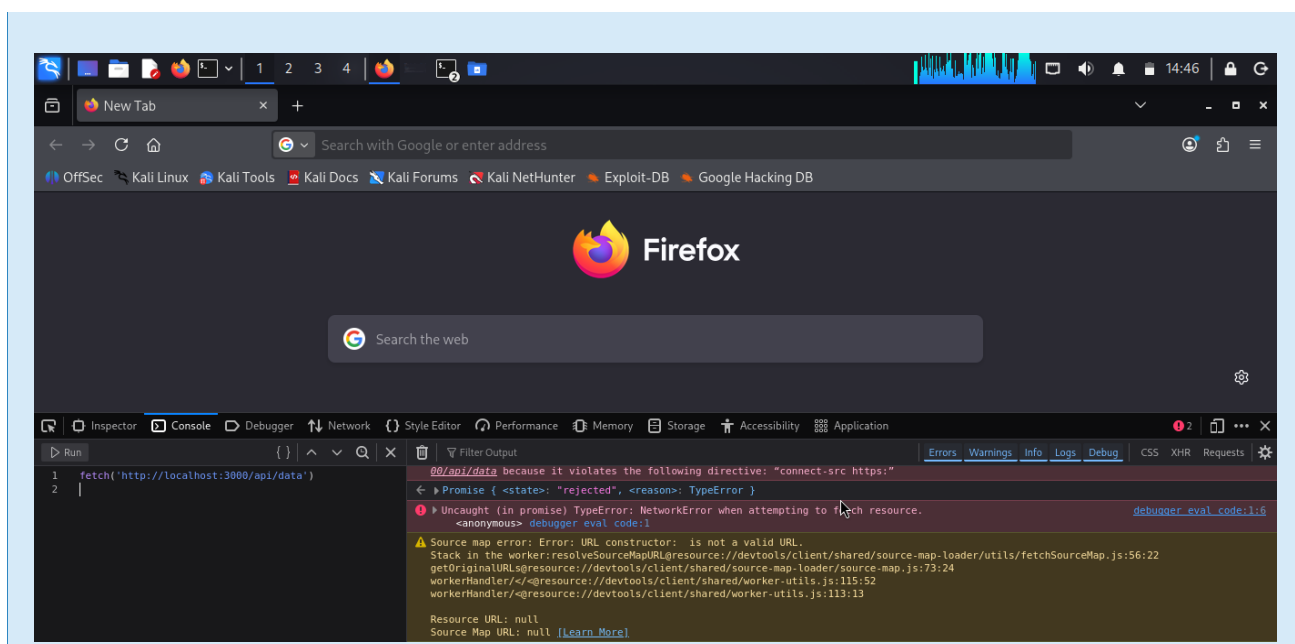
// .env file
API_KEY=your_super_secret_key_here
```



A terminal window titled 'suny@suny: ~/secure-api' displays the output of an API security hardening process. The first 15 lines are success messages: 'message': 'Success! API Security Hardening is working perfectly.' The next 5 lines are error messages: 'error': 'Too many requests, please try again later.'

```
suny@suny: ~/secure-api
{"message": "Success! API Security Hardening is working perfectly."}
{"message": "Success! API Security Hardening is working perfectly."}
{"message": "Success! API Security Hardening is working perfectly."}
{"message": "Success! API Security Hardening is working perfectly."}
{"message": "Success! API Security Hardening is working perfectly."}
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{"message": "Success! API Security Hardening is working perfectly."}
{"message": "Success! API Security Hardening is working perfectly."}
{"message": "Success! API Security Hardening is working perfectly."}
{"error": "Too many requests, please try again later."}
{"error": "Too many requests, please try again later."}
{"error": "Too many requests, please try again later."}
{"error": "Too many requests, please try again later."}
{"error": "Too many requests, please try again later."}
(suny@suny) ~/secure-api
$
```

Figure: API returning 401 Unauthorized when no API key / wrong key is provided



A Firefox browser window shows a 'New Tab' with the address bar set to 'http://localhost:3000/api/data'. The page content is a search bar. The developer console is open, showing a 'Run' button and a list of errors. The first error is a 'TypeError: NetworkError when attempting to fetch resource.' with a stack trace pointing to 'fetchSourceMap.js:56:22'.

```
1 fetch('http://localhost:3000/api/data')
2 |
// 00/api/data because it violates the following directive: "connect-src https:"
// Promise {<state>: "rejected", <reason>: TypeError }
// Uncaught (in promise) TypeError: NetworkError when attempting to fetch resource.
// Source map error: Error: URL constructor: is not a valid URL.
// Stack in the worker: resolveSourceMapURL@resource://devtools/client/shared/source-map-loader/worker-utils/fetchSourceMap.js:56:22
// getOriginalURL@resource://devtools/client/shared/source-map-loader/source-map.js:73:24
// workerHandler/c/@resource://devtools/client/shared/worker-utils.js:115:52
// workerHandler/@resource://devtools/client/shared/worker-utils.js:113:13
// Resource URL: null
// Source Map URL: null [Learn More]
```

Figure: API returning 200 OK when correct API key is provided in x-api-key header

Task 3: Security Headers & CSP Implementation

Objective: Implement Content Security Policy (CSP) to prevent script injections and enforce HTTPS using HSTS headers.

Step 1 — Install Helmet

Helmet automatically sets multiple security-related HTTP response headers:

```
npm install helmet
```

Step 2 — Enable Helmet (all default headers)

```
// index.js
const helmet = require('helmet');

// Enable all default security headers
app.use(helmet());

// Headers added by helmet() include:
// X-Content-Type-Options, X-Frame-Options, X-XSS-Protection,
// Referrer-Policy, Permissions-Policy, and more.
```

Step 3 — Add Content Security Policy (CSP)

CSP prevents XSS attacks by specifying which content sources are trusted:

```
app.use(
  helmet.contentSecurityPolicy({
    directives: {
      defaultSrc      : ["'self'"],
      scriptSrc       : ["'self'"],
      styleSrc        : ["'self'"],
      imgSrc          : ["'self'", "data:"],
      connectSrc      : ["'self'"],
      fontSrc         : ["'self'"],
      objectSrc       : ["'none'"],
      frameAncestors  : ["'none'"],
    },
  })
);
```

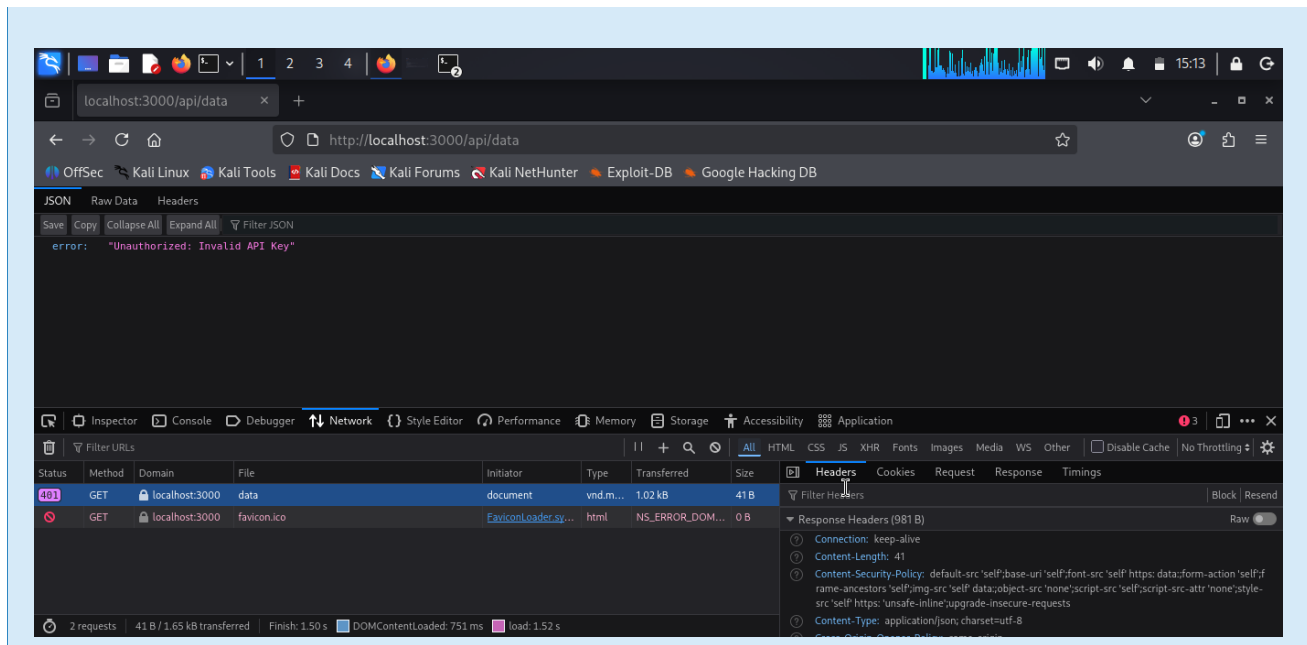


Figure: Browser DevTools Network tab showing Content-Security-Policy header in response

Step 4 — Enforce HTTPS with HSTS

HTTP Strict Transport Security (HSTS) forces browsers to always use HTTPS:

```
app.use(  
  helmet.hsts({  
    maxAge          : 31536000, // 1 year in seconds  
    includeSubDomains : true,  
    preload          : true,  
  })  
);
```

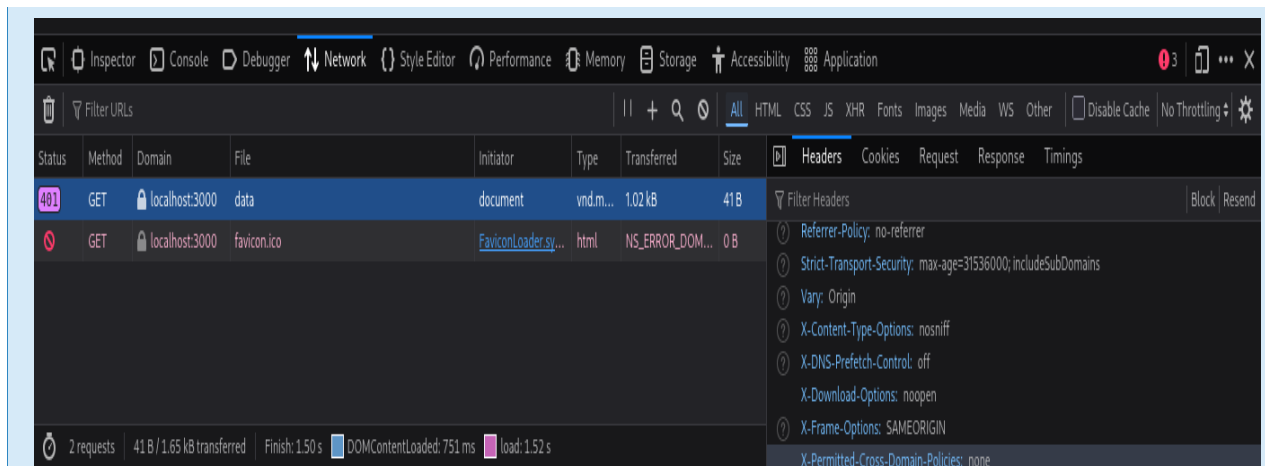


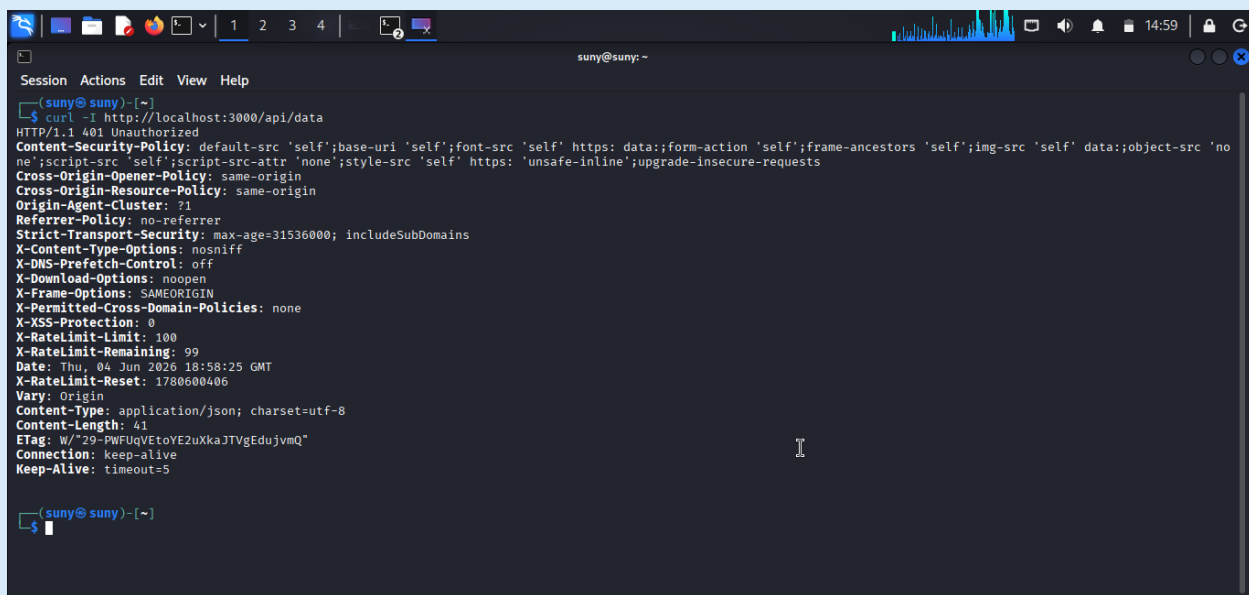
Figure: Browser DevTools showing Strict-Transport-Security header in HTTP response

Step 5 — Verify All Security Headers

Use a browser or curl to confirm all headers are correctly applied:

```
curl -I http://localhost:3000/api/test

# Expected headers in response:
# Content-Security-Policy: default-src 'self'; ...
# Strict-Transport-Security: max-age=31536000; includeSubDomains; preload
# X-Content-Type-Options: nosniff
# X-Frame-Options: SAMEORIGIN
# X-XSS-Protection: 0
```



```
suny@suny: ~
$ curl -I http://localhost:3000/api/data
HTTP/1.1 401 Unauthorized
Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action 'self';frame-ancestors 'self';img-src 'self' data:;object-src 'none';script-src 'self';script-src-attr 'none';style-src 'self' https: 'unsafe-inline';upgrade-insecure-requests
Cross-Origin-Opener-Policy: same-origin
Cross-Origin-Resource-Policy: same-origin
Origin-Agent-Cluster: ?1
Referrer-Policy: no-referrer
Strict-Transport-Security: max-age=31536000; includeSubDomains
X-Content-Type-Options: nosniff
X-DNS-Prefetch-Control: off
X-Download-Options: noopen
X-Frame-Options: SAMEORIGIN
X-Permitted-Cross-Domain-Policies: none
X-XSS-Protection: 0
X-Ratelimit-Limit: 100
X-Ratelimit-Remaining: 99
Date: Thu, 04 Jun 2026 18:58:25 GMT
X-Ratelimit-Reset: 1780600406
Vary: Origin
Content-Type: application/json; charset=utf-8
Content-Length: 41
ETag: W/"29-PWFUqVetoYE2uXkaJTVgEdujvmQ"
Connection: keep-alive
Keep-Alive: timeout=5

(suny@suny)~$
```

Figure: curl output showing all security headers applied correctly



```
(suny@suny)~$ nikto -h http://localhost:3000/api/data
- Nikto v2.6.0

+ Target IP: 127.0.0.1
+ Target Hostname: localhost
+ Target Port: 3000
+ Platform: Unknown
+ Start Time: 2026-06-04 15:18:55 (GMT-4)

+ Server: No banner retrieved
+ ERROR: Failed to check for updates: 403
+ [999100] /api/data/: Uncommon header(s) 'x-ratelimit-reset' found, with contents: 1780600808.
+ [999100] /api/data/: Uncommon header(s) 'origin-agent-cluster' found, with contents: ?1.
+ [999100] /api/data/: Uncommon header(s) 'x-ratelimit-limit' found, with contents: 100.
+ [999100] /api/data/: Uncommon header(s) 'x-ratelimit-remaining' found, with contents: 96.
+ No CGI Directories found (use '-C all' to force check all possible dirs). CGI tests skipped.
+ [013587] /api/data/: Suggested security header missing: permissions-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Permissions-Policy
```

Figure: Mozilla Observatory or securityheaders.com scan showing A+ grade

Security Implementation Summary

The table below summarises every security control implemented during Week 4:

Security Control	Tool / Method	Status	Evidence
Intrusion Detection	Fail2Ban	Implemented	Screenshot 1-2
Rate Limiting	express-rate-limit	Implemented	Screenshot 3
CORS Restriction	cors middleware	Implemented	Screenshot 4
API Key Authentication	.env + middleware	Implemented	Screenshot 5-6
Security Headers	Helmet.js	Implemented	Screenshot 7
Content Security Policy	helmet.contentSecurityPolicy()	Implemented	Screenshot 8
HSTS Enforcement	helmet.hsts()	Implemented	Screenshot 9-10

Challenges Faced & Lessons Learned

Challenges Faced	Lessons Learned
- Fail2Ban not detecting HTTP login attempts initially - CORS rejecting even valid same-origin requests during dev - CSP blocking inline scripts used in frontend - HSTS causing issues when testing over HTTP locally	- Custom filter rules in Fail2Ban need regex tuning per app - CORS origin whitelist must be exact including protocol - CSP nonces or hashes are needed for necessary inline scripts - HSTS should only be enabled after HTTPS is fully confirmed working

DELIVERABLES CHECKLIST

- ☒ Secured API with rate-limiting (express-rate-limit) — max 100 req / 15 min
- ☒ CORS configured to whitelist only trusted origins
- ☒ API key authentication middleware implemented and tested
- ☒ Security headers applied via Helmet.js
- ☒ Content Security Policy (CSP) configured to prevent script injection
- ☒ HSTS header enforcing HTTPS for 1 year
- ☒ GitHub repository updated with code and detailed README.md
- ☒ All screenshots attached as evidence for each implementation step

Conclusion

All three Week 4 tasks have been successfully completed. The application is now protected against brute-force attacks through Fail2Ban and rate limiting, unauthorized API access through API key authentication, cross-origin attacks through CORS configuration, and script injection through CSP and HSTS headers. The implementation follows OWASP best practices and prepares the groundwork for the ethical hacking exercises in Week 5.

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